

Rainstorm 2021

Special Relativity & Black Holes

Table of Contents

Part 1: Special Relativity

- 1-2 Twin Paradox
- 3-5 Pythagorean Theorem
- 6-7 Einstein's 4D spacetime
- 8-9 Speed of light is constant
- 10-11 Time dilation

Part 2: General Relativity

- 12-14 Gravity is curved spacetime
- 15 Triangles on curved surfaces
- 16-17 Expansion of our universe/ Big Bang
- 18-19 Math of mass curving space (gravity=geometry) -
- 20-23 Black Holes
- 24-25 Black Hole radiation (Hawking radiation)
- 26-27 Quantum Gravity / String theory
- 28-35 Shape of the Universe ✓

Link to recorded classes: https://www.youtube.com/playlist?list=PLmW_iomNe2Qg0WxOB_Ok1_P1NpIPeqQBU

Or in: <https://www.youtube.com/channel/UC5AF9LTrXRCf3aDfGhgmHCQ>

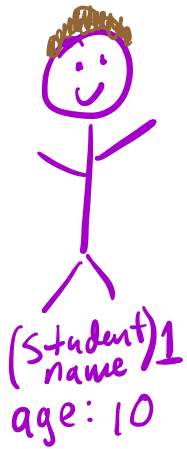
1.) Whats special about Special relativity?

①

1905

Twin Paradox (conceptual)

2021



(Student) 2
age: 10

$\approx 18,000,000$
mph

Comes home 50 years later....

2071



(Student) 1
age: 60



(Student) 2
age: 40

Q: Why does this happen?

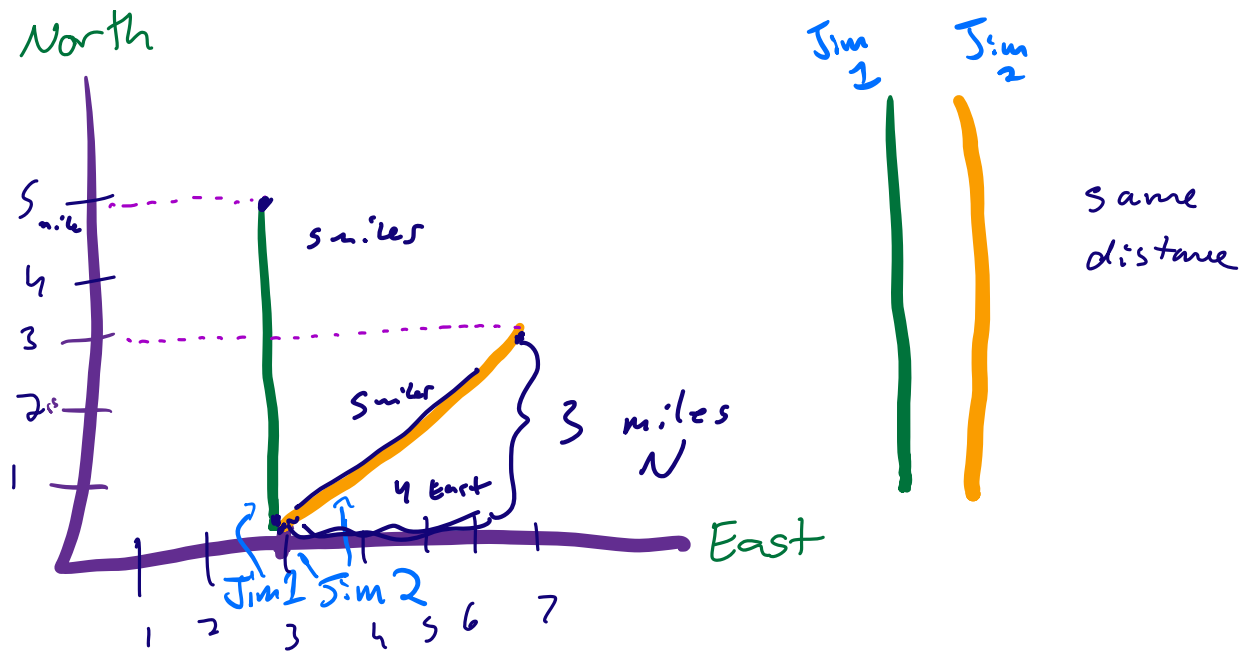
Because the Universe is 4D!
4th Dimension is time.

3D we see:
Space
up/down
forward/backward
left/right
+
Time! (1D)
= 4D Spacetime

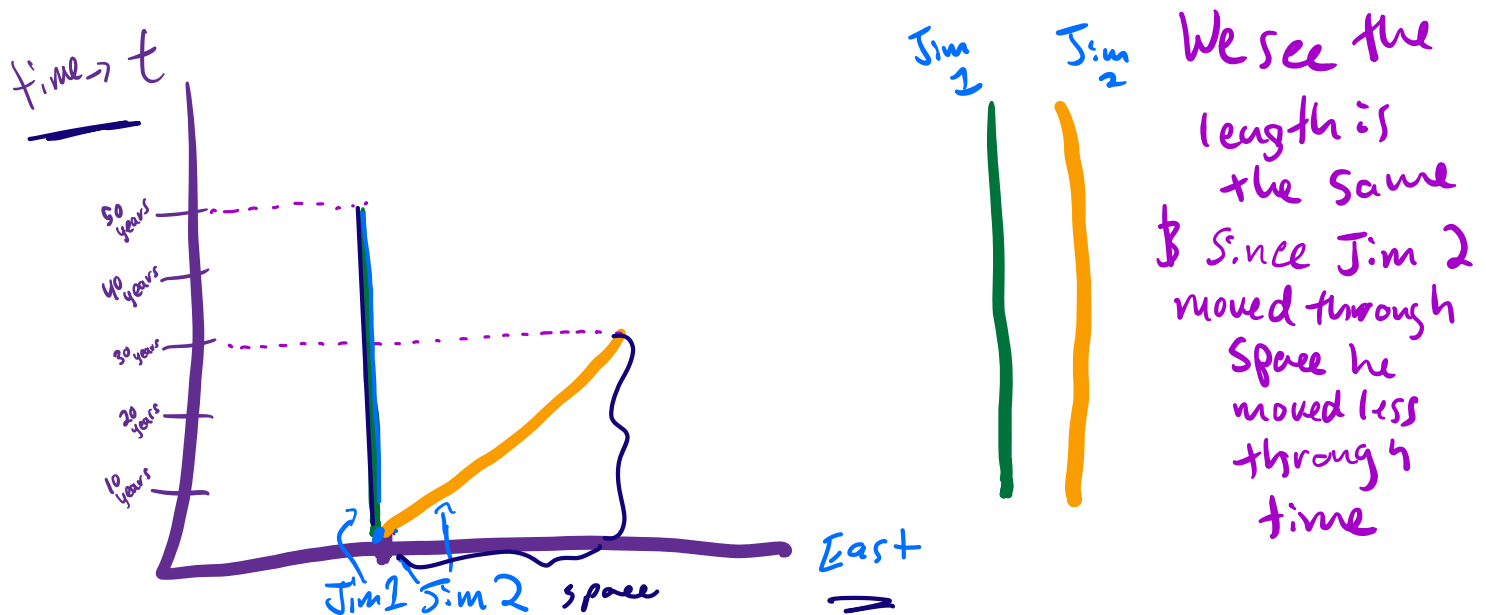


We treat time as another direction to move in.

The faster one moves through space (2)
the slower one moves through time



(Slightly inaccurate diagram but good for current purpose)



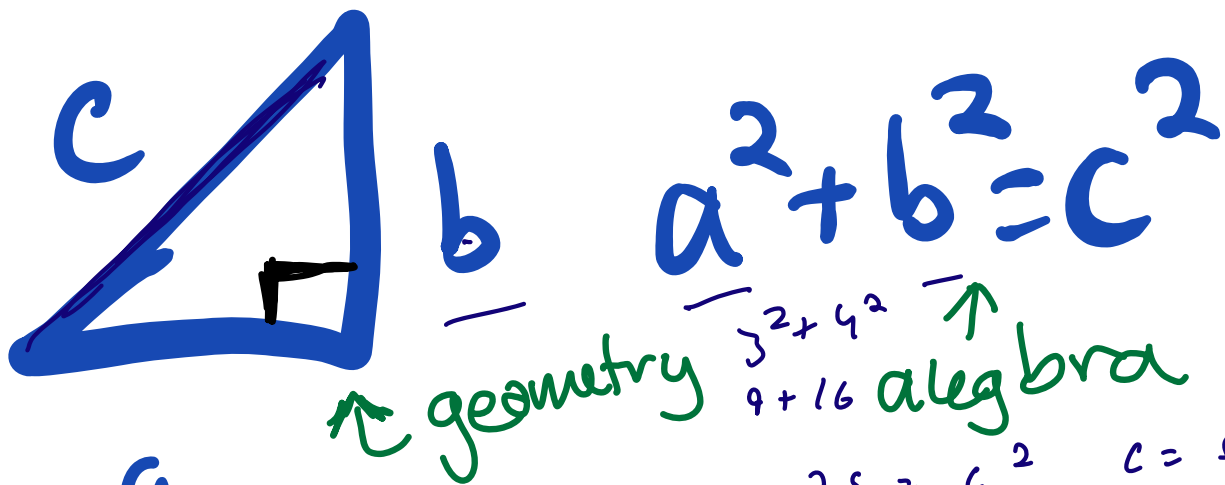
At low speeds ($\approx 60 \text{ mph}$) this still happens but the effect is negligible

The next question is why the Universe is 4D, well get to that, but first let's understand this 4D world.

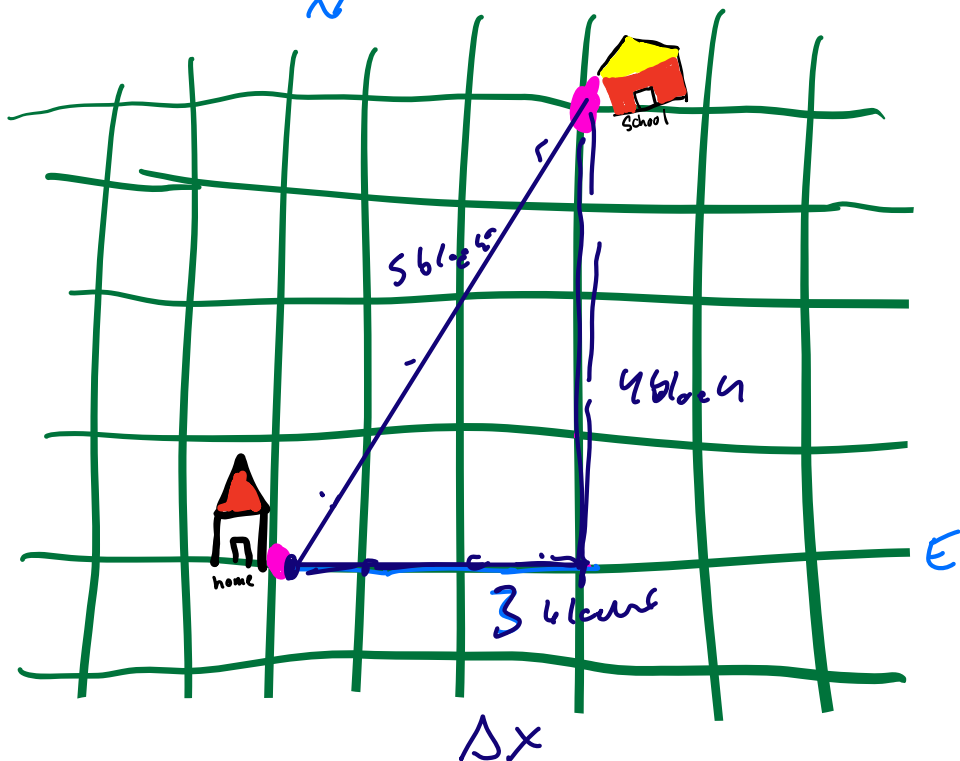
2.) How do we make a 4 dimensional geometry that involves a time dilation?

③

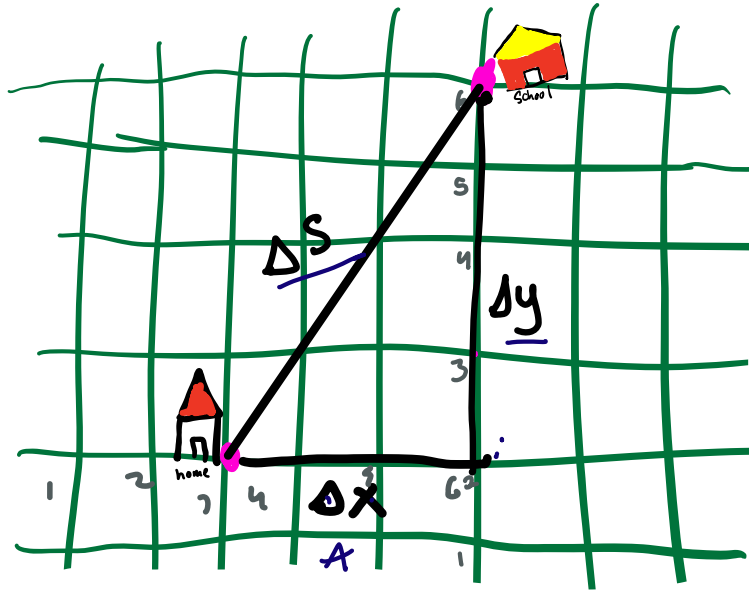
First lets review :



We can use this every day



We call the shortest distance between home and school " ΔS "



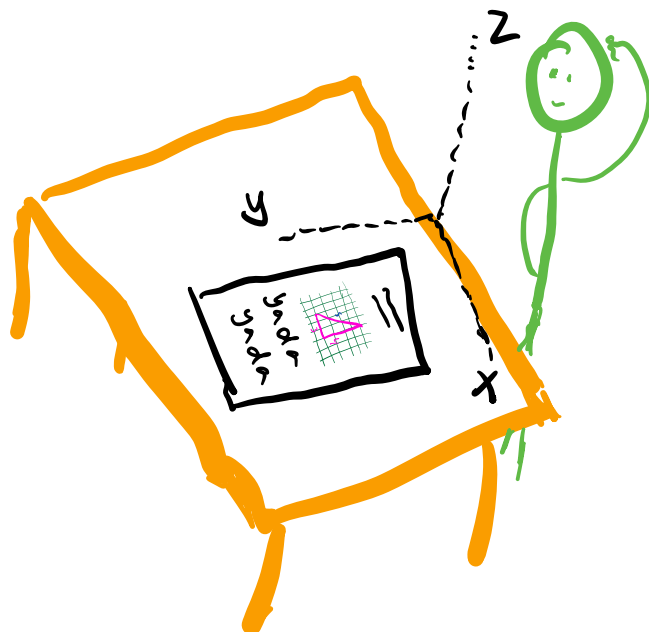
Using $a^2 + b^2 = c^2$ we find $3^2 + 4^2 = 5^2$!

In the 2D paper world we say

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2$$

↑
east/west
↑
north/south

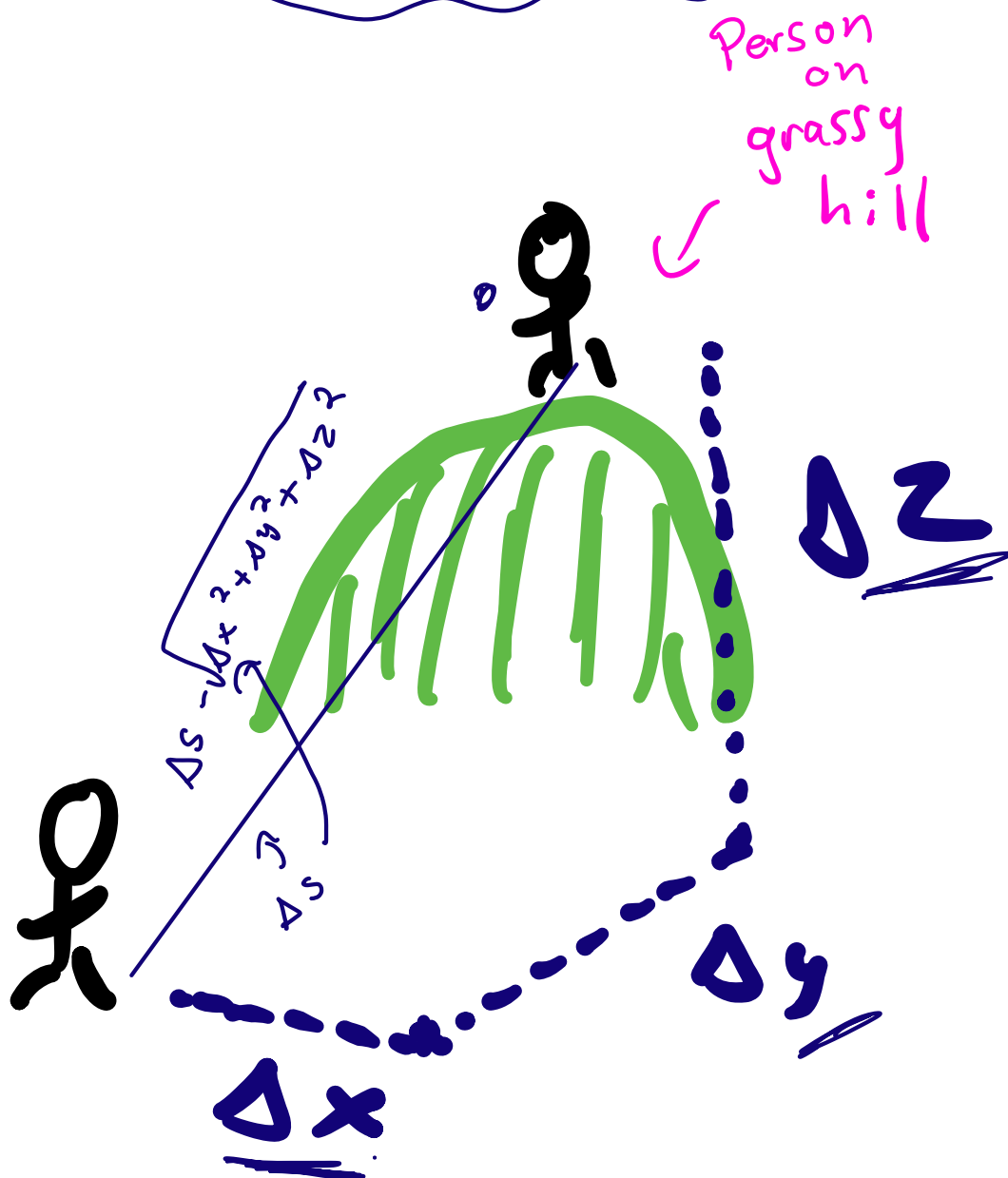
But we do not live on paper...



The world our eyes see:

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

$\uparrow$ $\uparrow$ $\uparrow$
E/W N/S up/down



6

But wait! Didn't all this twin paradox business tell us we live in 4 dimensions?

Naively we will write

$$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 + (\overset{\text{time}}{\downarrow} \Delta t)^2$$

Einstein showed the geometry of our universe is:

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c \Delta t)^2 \quad \leftarrow$$

"miles" "miles" "miles" ↑
minus
sign!

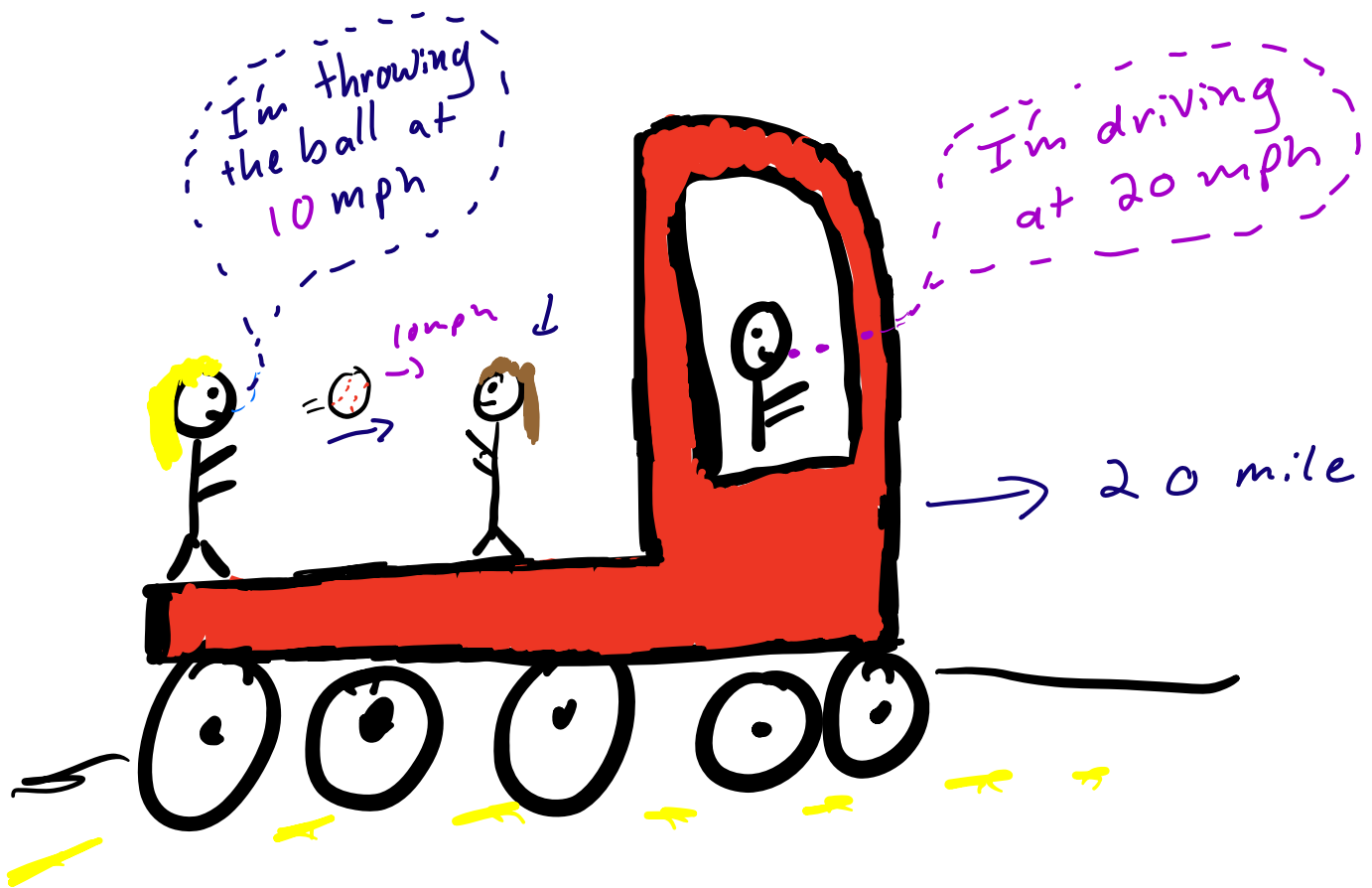
$$c = 6.07 \times 10^8 \frac{\text{miles}}{\text{hr}}$$

$$c \Delta t = \frac{6.07 \times 10^8 \text{ miles}}{\text{hr}} \times \text{hr}$$

3.) Why the speed of light? ⑧

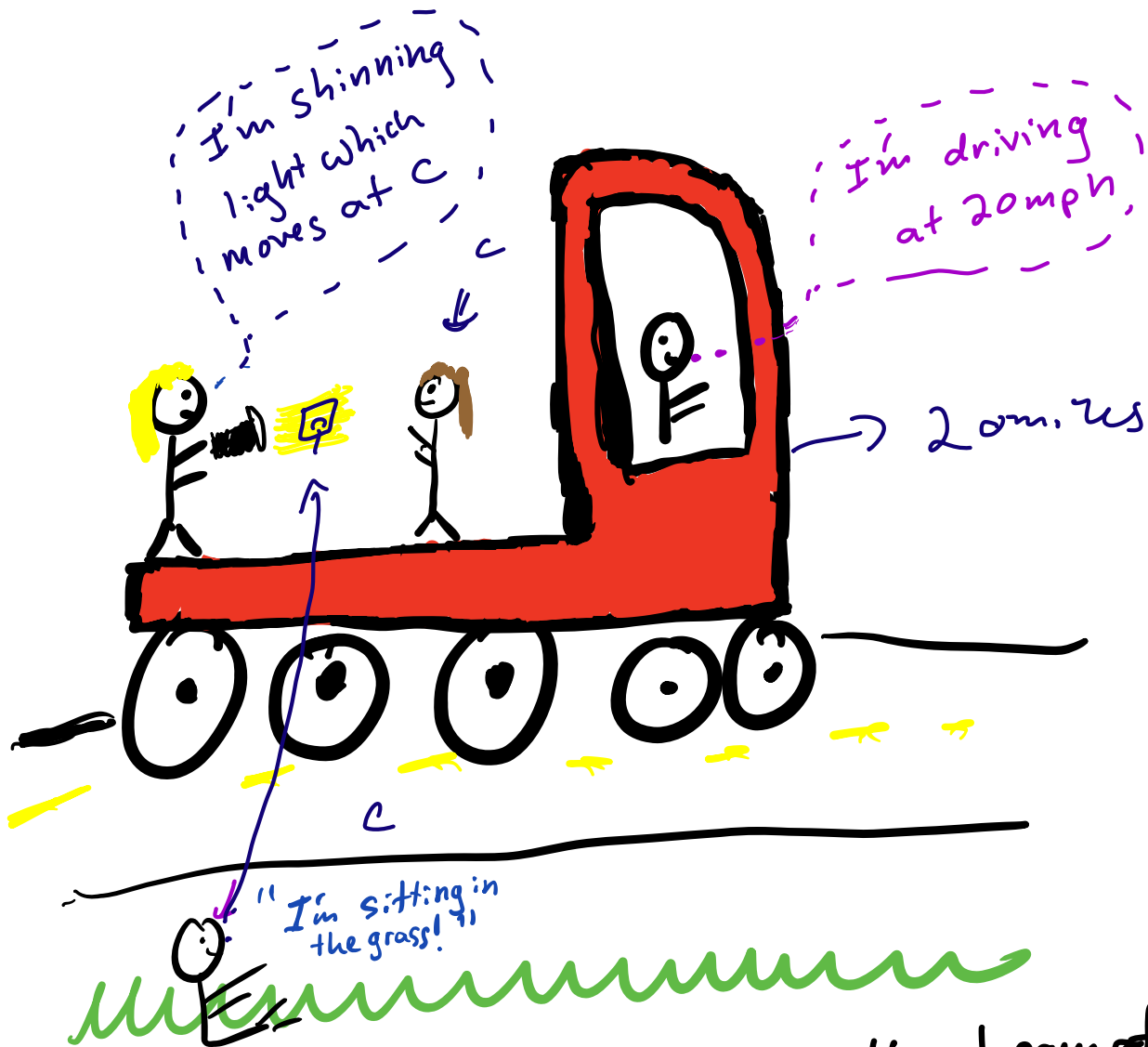
According to the laws of electricity & magnetism light moves the same speed for everyone! (see Michelson-Morley Experiment)

Let's understand this fascinating fact



Q: what speed does the man see the ball move?

$$10 \text{ mph} + 20 \text{ mph} = 30 \text{ mph}$$



Q: What speed does the man see the beam of light move? $C + 20\text{mph} = C$?

This is obvious not the right way to look at things. It turns out for this experimentally observed principle to hold we must live in

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c\Delta t)^2$$

$$d = S \cdot t \quad \frac{\Delta x}{\Delta t} = \frac{5 \text{ mile east}}{1 \text{ hour}} = 5 \text{ mph} = V_x$$

$$\frac{\Delta y}{\Delta t} = V_y \quad \frac{\Delta z}{\Delta t} = V_z$$

A consequence of this is that
time dilates, length contracts,
and nothing can go faster than light

$$\frac{(\Delta S)^2}{(c\Delta t)^2} = \frac{(\Delta x)^2}{(c\Delta t)^2} + \frac{(\Delta y)^2}{(c\Delta t)^2} + \frac{(\Delta z)^2}{(c\Delta t)^2} - \frac{(c\Delta t)^2}{(c\Delta t)^2}$$

$$\frac{(\Delta S)^2}{c^2(\Delta t)^2} = \frac{v_x^2}{c^2} + \frac{v_y^2}{c^2} + \frac{v_z^2}{c^2} - 1$$

$$= \frac{v_x^2 + v_y^2 + v_z^2}{c^2} - 1 = \frac{v^2}{c^2} - 1$$

$$-(\Delta t)^2 \cdot \frac{-(\Delta z)^2}{(\Delta t)^2} = \left(\frac{v^2}{c^2} - 1 \right) \cdot 1 \cdot \Delta t^2$$

Speed of motion

$$(\Delta \tau)^2 = (\Delta t)^2 \left(1 - \frac{v^2}{c^2} \right)$$

time of mover time of person

$$(\Delta \tau)^2 = - \frac{\Delta S^2}{c^2}$$

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c\Delta \tau)^2$$

$$(\Delta S)^2 = -c^2(\Delta \tau)^2$$

What space time distance does light move

$$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - c^2 (\Delta t)^2$$

$$= (\Delta t)^2 \left(\underbrace{\left(\frac{\Delta x}{\Delta t} \right)^2 + \left(\frac{\Delta y}{\Delta t} \right)^2 + \left(\frac{\Delta z}{\Delta t} \right)^2}_{v_x^2 + v_y^2 + v_z^2 = v^2} - c^2 \right)$$

$$(\Delta t)^2 (v^2 - c^2) = (\Delta s)^2$$

$v = c$

$$(\Delta t)^2 (\underbrace{c^2 - c^2}_0) = (\Delta s)^2$$

$$(\Delta \tau)^2 = (\Delta t)^2 \left(1 - \frac{v^2}{c^2} \right) =$$

$v \approx .99999 c$

$$(\Delta \tau)^2 = (\Delta t)^2 \left(1 - \frac{v^2}{c^2} \right)$$

$\uparrow$
time of mover
 $\uparrow$
time of person
 $\downarrow$
speed of motion

If I wait for the premier
of Frozen 3 for 10 hours, how
long will my twin jealously watch
me wait in line if he is moving
 $\frac{1}{2}$ the speed of light

$$\Delta t = 10 \text{ hours}$$

$$v = \frac{1}{2}c \quad (\Delta \tau)?$$

$$\left(\overset{\substack{\uparrow \\ \text{time of} \\ \text{mover}}}{\Delta \tau} \right)^2 = \left(\underset{\substack{\uparrow \\ \text{time of} \\ \text{person}}}{\Delta t} \right)^2 \left(1 - \overset{\substack{\downarrow \\ \text{speed of} \\ \text{mover}}}{\frac{v^2}{c^2}} \right)$$

$$\left(\frac{1}{2}c \right)^2 = \frac{c^2}{2^2} = \frac{c^2}{4}$$

$$= (10)^2 \left(1 - \frac{\left(\frac{1}{2}c \right)^2}{c^2} \right) =$$

$$(\Delta \tau)^2 = 75$$

$$\Delta \tau = 8.66 \text{ hours}$$

$$= 100 \left(1 - \frac{1}{4} \frac{c^2}{c^2} \right) = 100 \left(\frac{4}{4} - \frac{1}{4} \right)$$

$$= 100 \left(\frac{3}{4} \right)$$

$$= 25 \cdot 3 = 75$$

General Relativity

(12)

What is gravity?

We learn in school Newton discovered that "mass attracts mass"

The equation is $F = \frac{G m_1 m_2}{r^2}$

Force moving them together.

two massive object

light

$$(\Delta \tau)^2 = (\Delta t)^2 \left(1 - \frac{v^2}{c^2} \right)$$

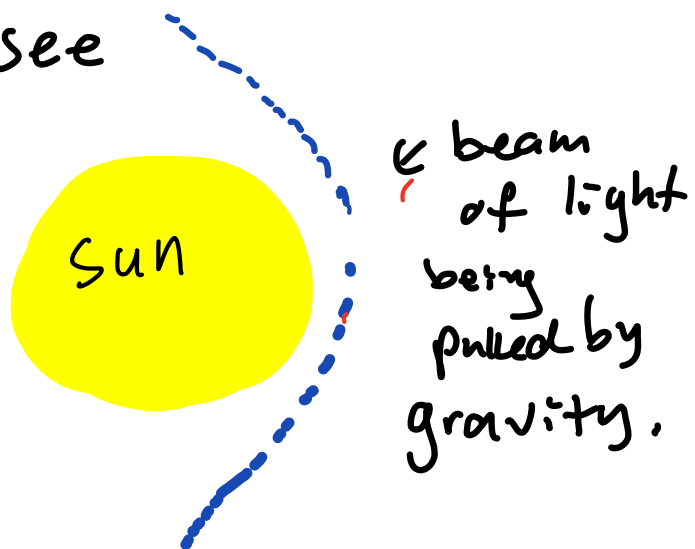
However in 1917 an experiment was done
to show this is wrong!

Light has no mass but yet gets attracted by gravity

Newton said:

$$F = \frac{G m_{\text{sun}} 0}{r^2} = 0$$

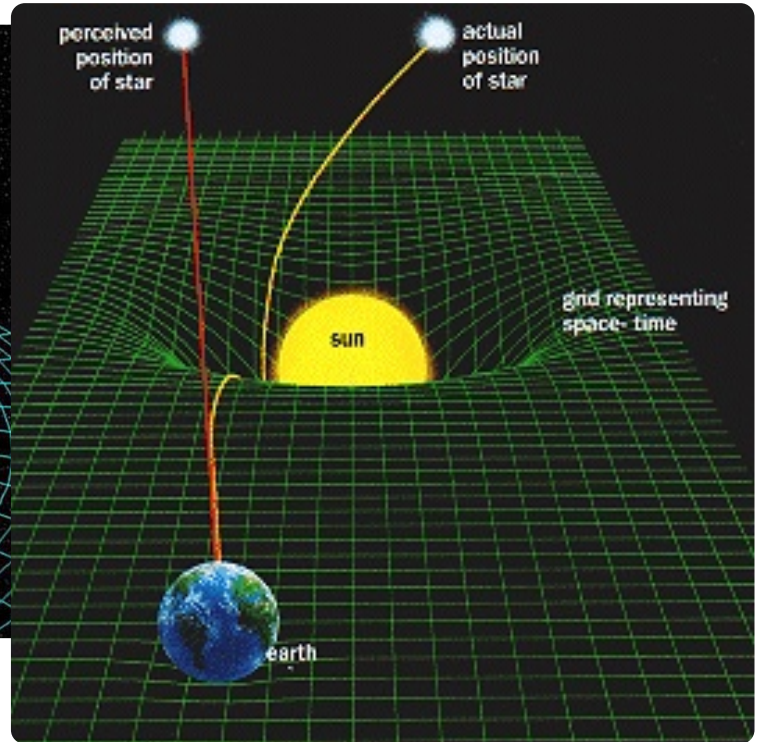
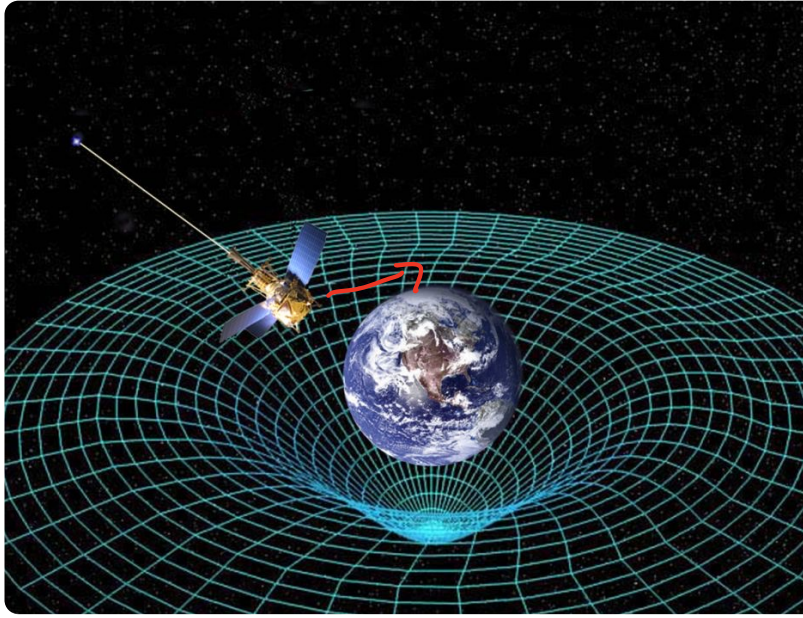
We see



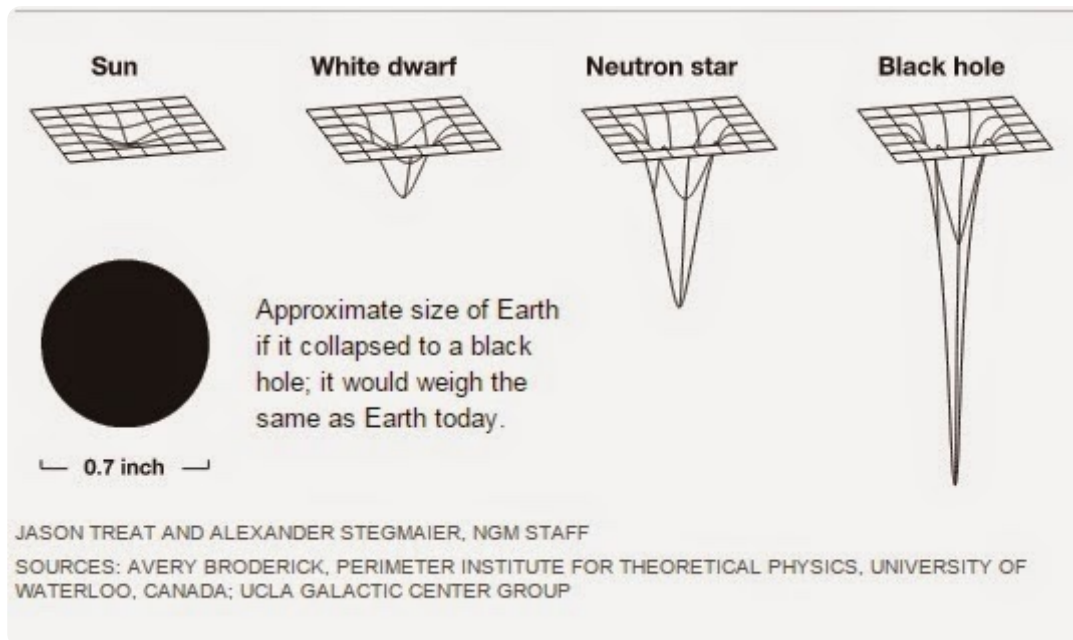
Einstein concluded that what we gravity is actually 4D spacetime that's curved. (13)

Mass Causes Space time to bend.

The more heavy an object the more it bends space



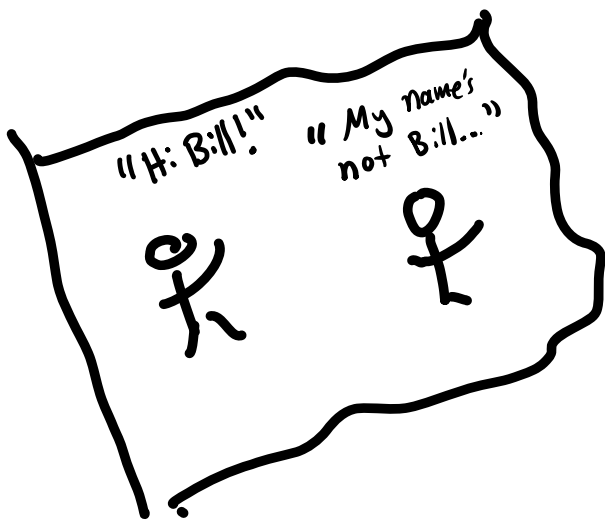
The smaller the space with the same mass Curves space time more.



We don't see space curved.

This is because we only see in 3D

An Example: Say we live in the 2D paper world.

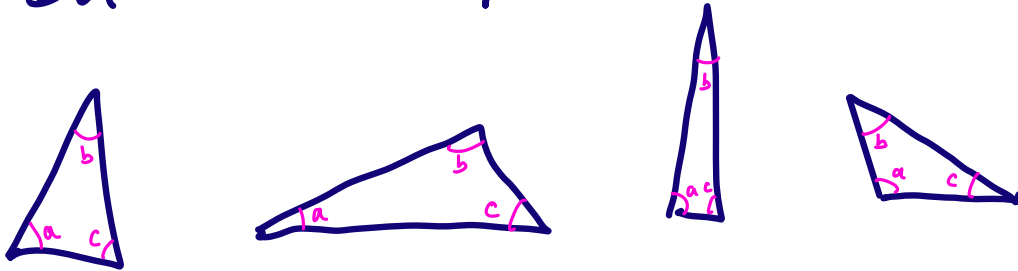


If we bend/curve the paper in the 3rd dimension the paper people wouldn't notice.

However we can use geometry to show our space is intrinsically curved.

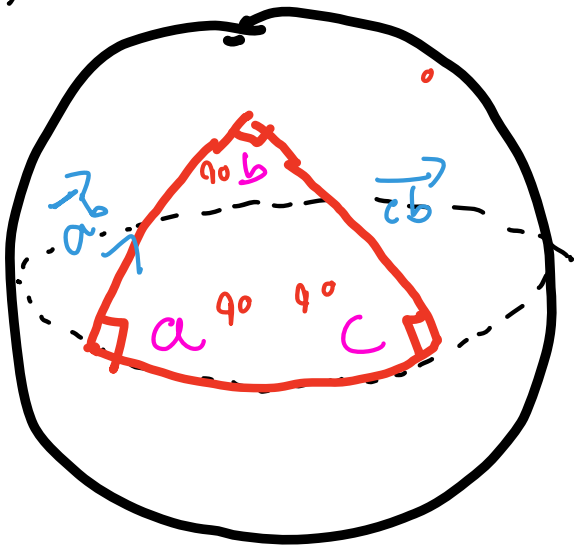
One property of curved surfaces is 15
that the sum of angles of a triangle
 $\neq 180^\circ$

In flat space



For all of these $a + b + c = 180^\circ$

Now take a ball: I can draw 3
 90° angles and
get a triangle



$$a + b + c = 270!$$

Try it yourself!

This also means two parallel lines
will intersect.

We can apply these properties to
objects falling in gravity to calculate the
curvature of space.

Since the Sum of the angles $\neq 180$
Pythagorean theorem fails

How to modify:

$$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - c^2 \Delta t^2 ?$$

Expanding universe:

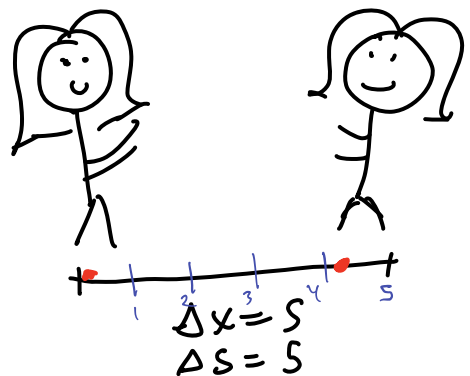
$$\underline{(\Delta s)^2} = a(t) \left((\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 \right) - c^2 \Delta t^2$$

$$a(t) = t$$

$$(\Delta s)^2 = (\Delta x)^2$$

$\Delta x = 5 \quad \Delta s = 5$

Let two friends be
 separated by $\Delta x = 5$ at
 $t = 1, a(t) = 1$



Their distance at $t=1$:

$$(\Delta s)^2 = 1 \cdot (\Delta x)^2 = 1 \cdot (5)^2 = 25$$

$$\Delta s = 5$$

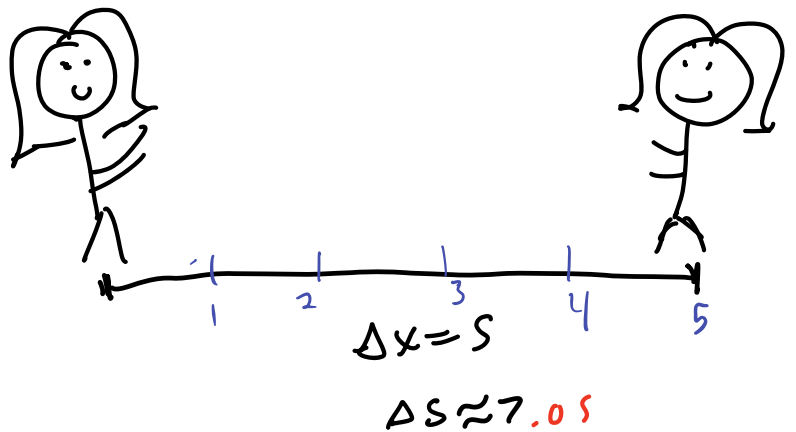
at $t=2$: $(\Delta s)^2 = 2(\overset{5 \text{ blocks}}{\Delta x})^2 + (\overset{0}{\Delta y})^2 + (\overset{0}{\Delta z})^2 - (c \overset{0}{\Delta t})^2$

$a(2) = 2$ Without moving the space between the twins grew.

$$(\Delta s)^2 = 2(\Delta x)^2$$

$$= 2 \cdot (5)^2 = 50$$

$$\sqrt{(\Delta s)^2} = \sqrt{50} \Rightarrow \Delta s = 7.07$$



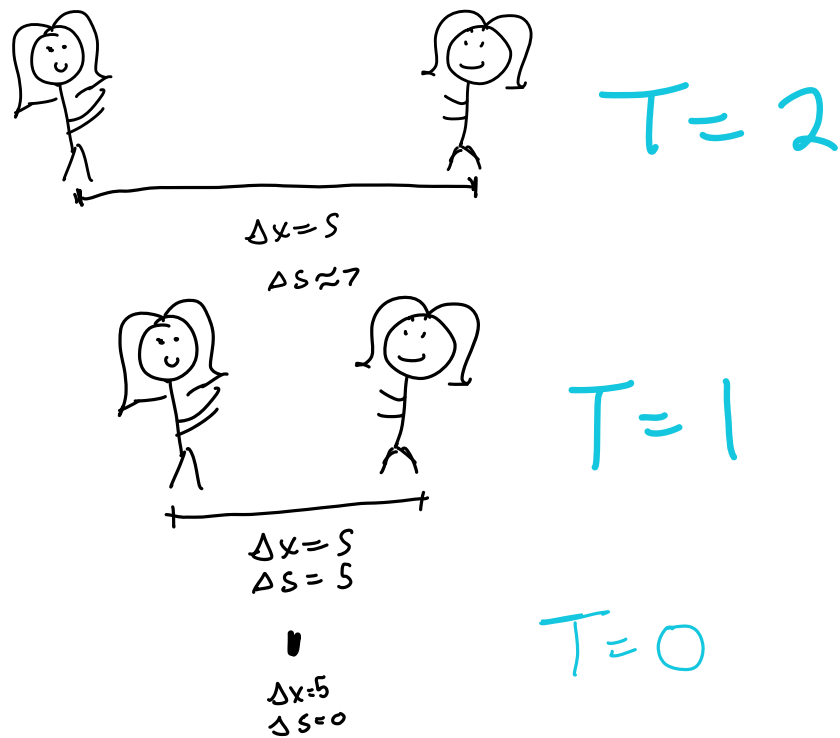
Big Bang.

17

Take $a(t) = t$ at $t = 0$

$$(\Delta S)^2 = 0 \left((\overset{s}{\Delta x})^2 + (\overset{a}{\Delta y})^2 + (\overset{a}{\Delta z})^2 \right) - (c \overset{a}{\Delta t})^2$$

This model suggests the distance between the friends at $t = 0$ is $\Delta S = 0$



Thus, we found a geometry that includes the big bang. We considered $a(t) = t$. Our universe is not just expanding it is accelerating. If you have taken highschool physics than you'll know that $a(t) \sim t^2$ gives acceleration.

Now that we seen a possible correction (18)
 Our "distance equation" (aka Metric):

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c\Delta t)^2$$

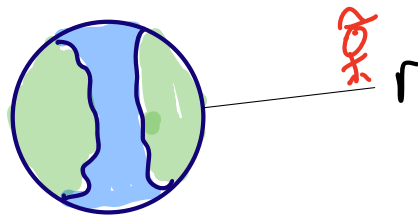
Lets see how we can change it to
 include the effects of gravity:

roughly looks like:
 When $\Delta x, \Delta y, \Delta z, \Delta t \rightarrow 0$:

$$V = \frac{GM}{r}$$

$$(\Delta S)^2 = \frac{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}{\left(1 - \frac{GM}{c^2 r}\right)} - (c\Delta t)^2$$

This is a curved metric



$$(\Delta S)^2 = -\left(1 - \frac{GM}{c^2 r}\right) (c\Delta t)^2 + \frac{(\Delta r)^2}{\left(1 - \frac{GM}{c^2 r}\right)} + r^2 (\Delta\theta^2 + \sin^2\theta (\Delta\phi)^2)$$

Where on a sphere: $x = r \sin\theta \cos\phi$ $y = r \sin\theta \sin\phi$ $z = r \cos\theta$

$$(\Delta S)^2 = \frac{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}{\left(1 - \frac{GM}{c^2 r}\right)} - \left(1 - \frac{GM}{c^2 r}\right) (c \Delta t)^2$$

$$r \rightarrow \infty \quad \lim_{r \rightarrow \infty} \left(1 - \frac{GM}{c^2 r}\right) = \left(1 - \frac{GM}{c^2 \infty}\right) = (1-0) = 1$$

So in this limit:

$$(\Delta S)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 - (c \Delta t)^2$$

Some thing curious happens when $r \rightarrow \frac{GM}{c^2}$

$$\lim_{r \rightarrow \frac{GM}{c^2}} \left(1 - \frac{GM}{c^2 r}\right) = \left(1 - \frac{GM}{c^2 \frac{GM}{c^2}}\right) = 1 - 1 = 0$$

$$\Delta z = \frac{\Delta S^2}{c^2}$$

$$R = \frac{GM}{c^2}$$

$$\star (\Delta S)^2 = \frac{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}{\left(1 - \frac{GM}{c^2 r}\right) = 0} - \left(1 - \frac{GM}{c^2 r}\right) (c \Delta t)^2$$

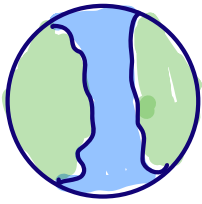
$$\left(1 - \frac{GM}{c^2} \frac{1}{\frac{GM}{c^2}}\right) = (1-1) = 0$$

When all the mass of a star

Collapses into a star of radius

$R = \frac{GM}{c^2}$, it forms a **BLACK HOLE**
↑
called event horizon

If the earth can form a black hole
by how much must it shrink?



$$R_{\text{event horizon}} = \frac{GM}{c^2}$$

$$\text{Mass} \approx \frac{5.97 \times 10^{24} \text{ kg}}{1}$$

$$G \approx 6.67 \times 10^{-11} \frac{\text{m}^3}{\text{kg s}^2}$$

$$c \approx 3 \times 10^8 \frac{\text{m}}{\text{s}}$$

$$R_{\text{event horizon}} = .4 \text{ cm}$$

↑
 $\frac{1}{4}$ the size
of dime

$$R_{\text{radius of Earth}} = 3,958 \text{ miles}$$

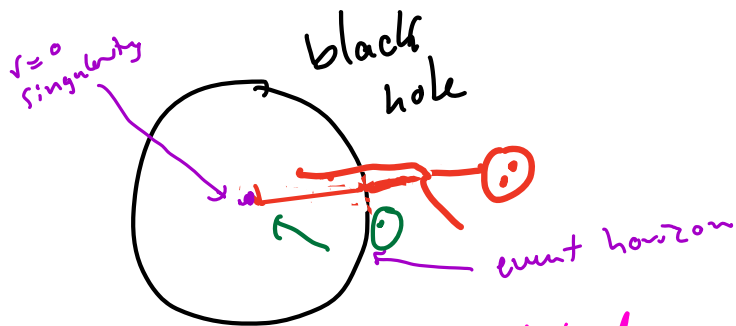
$$\text{Mass} = 75 \text{ kg}$$

$$r_{\text{event horizon}} = 5.5 \times 10^{-26} \text{ cm}$$

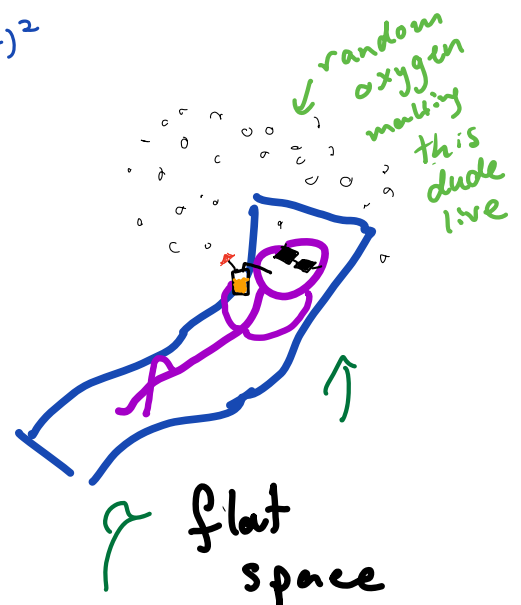


As stated, someone looking at the black hole from the outside will see time "stops" at the event horizon and will never see the person cross.

$$(\Delta s)^2 = 0(\Delta t)^2 + \infty(\Delta r)^2$$



Poor person being stretched entering a black hole since gravity is stronger at the feet than the head.



The person falling into a black hole will do so in a short amount of time
 ~ 6 microseconds to go from horizon to singularity.

In the fraction of a second it takes
for the person to fall into the black hole,

(22)

INFINITE amount of time passes for the
person chillin outside.

This is one of the reasons why **Nothing**
Can escape a black hole!

In order to escape a black hole you
must go faster than light. Which
requires an infinite amount of energy and
is not possible.

Light cannot escape a black hole
We can see this with a diagram.
In flat space light moves at 45°



Why 45°
well in flat space
light only moving east is

$$1) \Delta s^2 = \Delta x^2 - c^2 \Delta t^2$$

$$2) \Delta s^2 = \Delta t^2 \left(\frac{\Delta x^2}{\Delta t^2} - c^2 \right)$$

$$\text{speed is } \frac{\Delta x}{\Delta t} = c$$

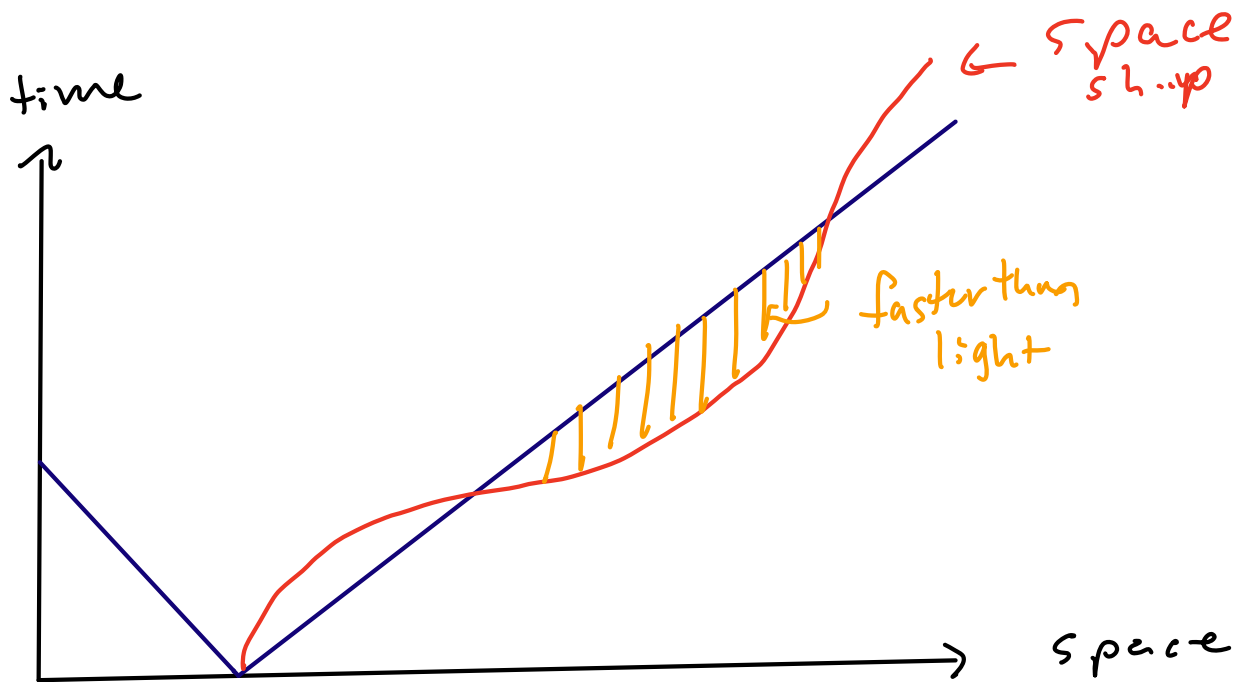
$$= \Delta t^2 (c^2 - c^2) = 0$$

$$\Delta s^2 = 0$$

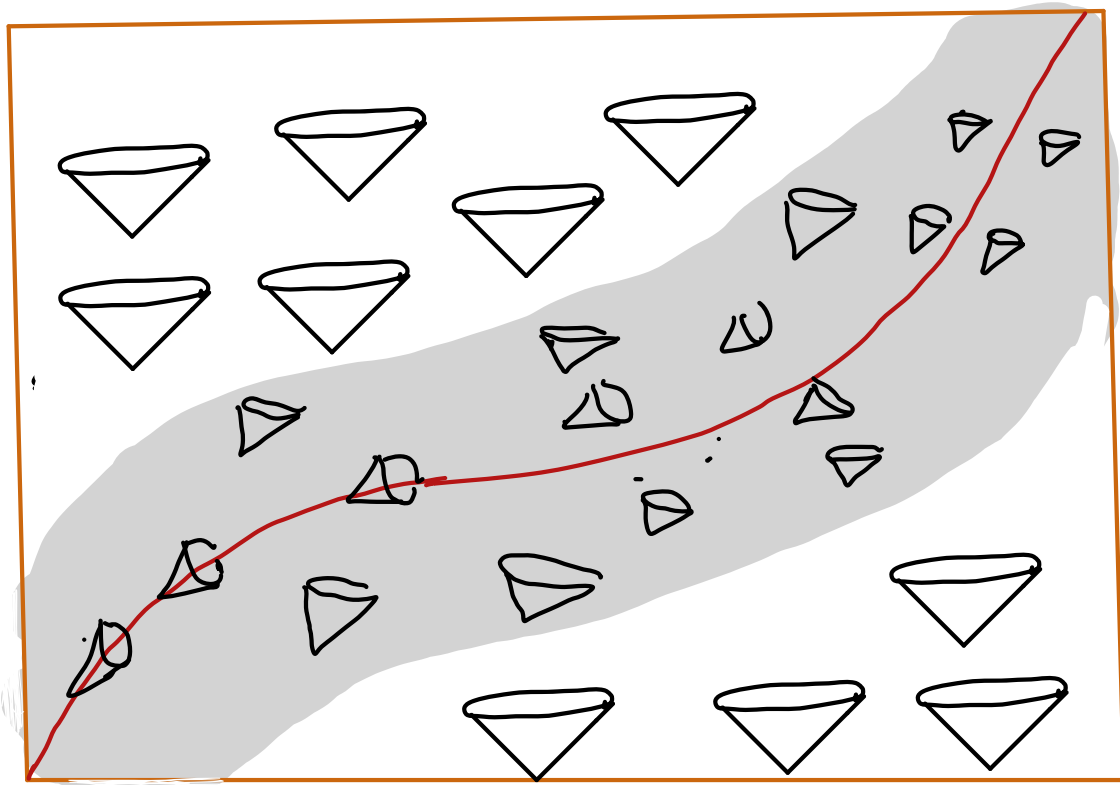
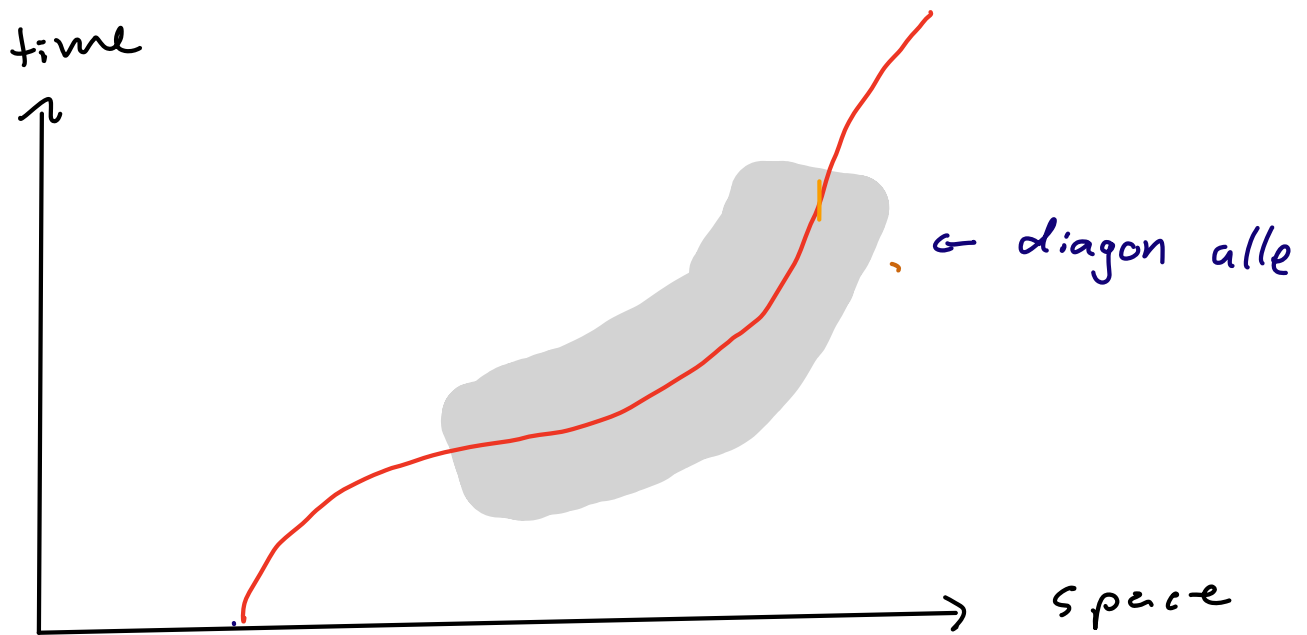
$$\Rightarrow \Delta x = c \Delta t$$



Traveling faster than light?

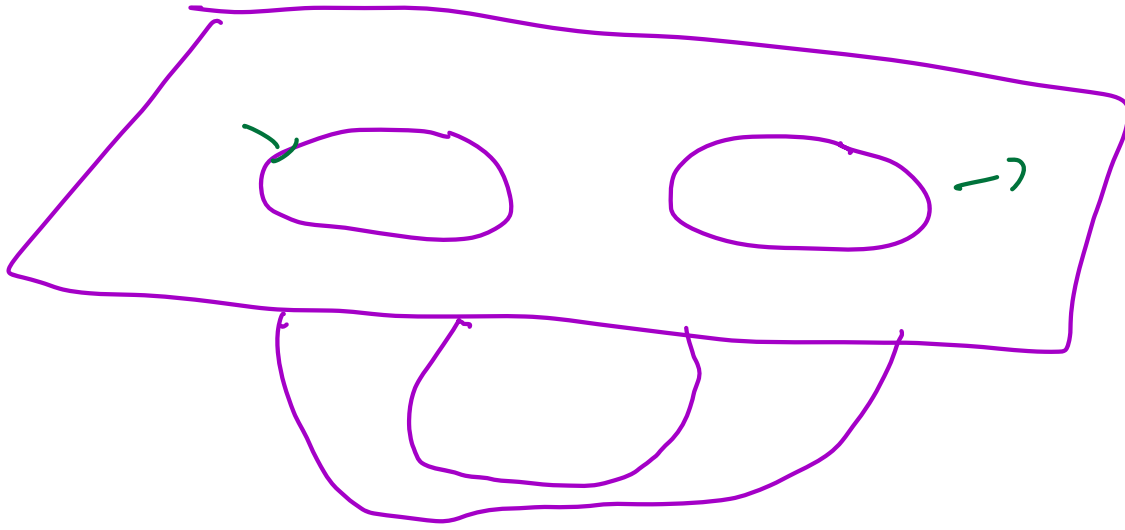


Is this possible?



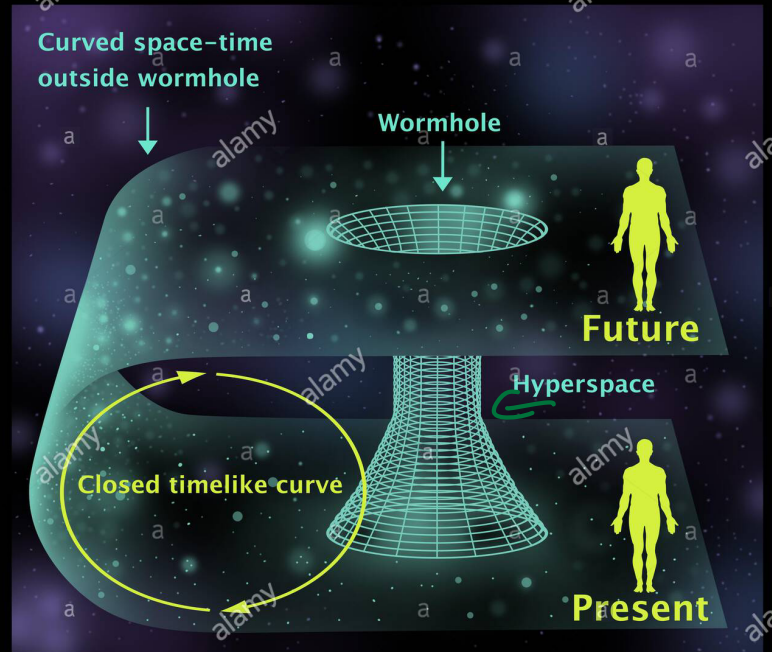
$$(\Delta s)^2 \approx -(c\Delta t)^2 + [\Delta x - v_s(t) f(r_s) \Delta t]^2 + (\Delta y)^2 + (\Delta z)^2$$

Worm holes



↑
Throat of wormhole

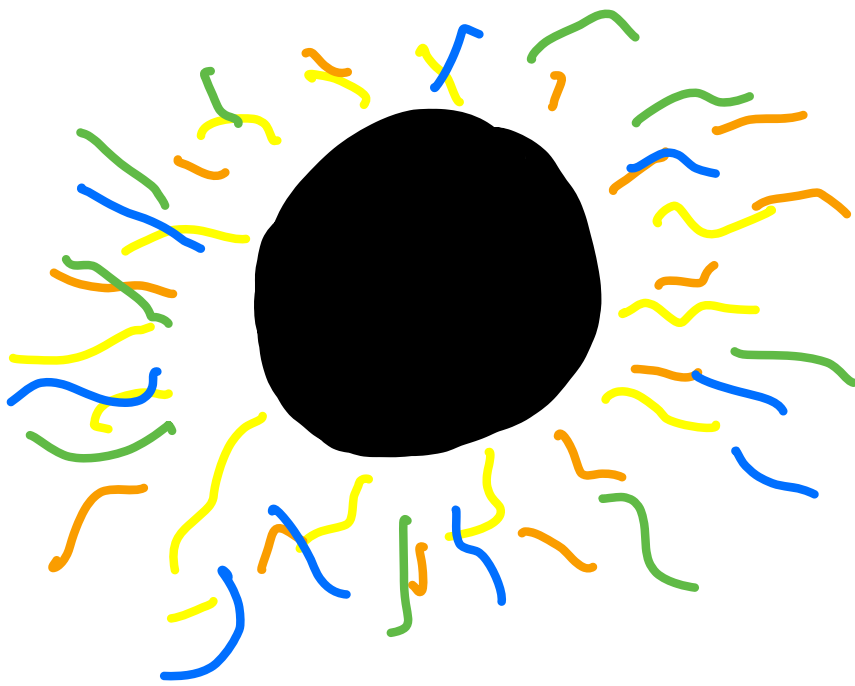
Wormholes



Hawking Radiation

27

Black holes, just like everything else, radiate.

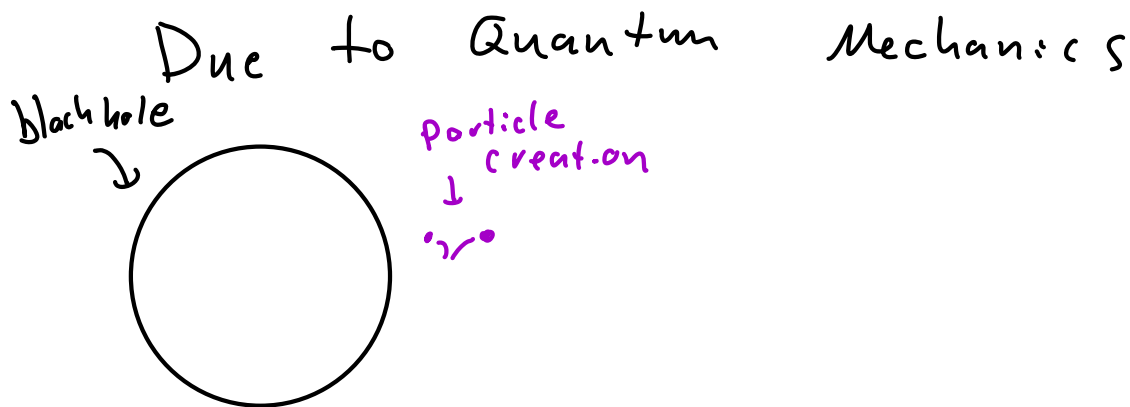


It can radiate all different things

EX:

- Light
- Electrons
- parts of protons (quarks)

This is bizarre considering a black hole is just space itself.

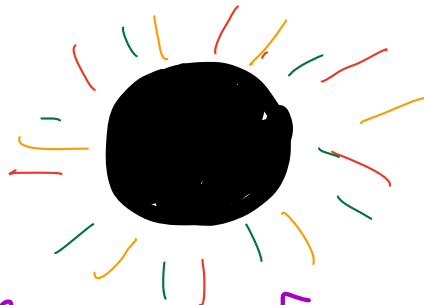
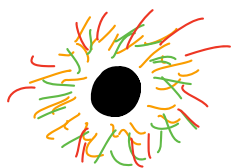


The hotter things are the more they radiate.

The temperature of a black hole is.

temp $\rightarrow T = \frac{1}{8\pi M} \leftarrow \text{mass}$

Therefore big massive black holes are colder and radiate less than lighter smaller black holes.



Smaller blackhole loses mass faster than a bigger more stable one.

The time it will take for a black hole the mass of the earth to evaporate is 10^{67} years.
One time of kanye is 2.2×10^{-40} seconds
Big Difference!

Can a black hole radiate away to nothing?

$10^{-35} \text{ m} \leftarrow$

Quantum Gravity / String Theory (26)

The issue with answering this question is we don't know what space means at distances less than $3 \times 10^{-35} \text{ m}$.

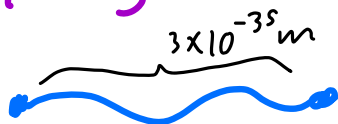
Using Einstein's theory there is no way to talk about a black hole smaller than $3 \times 10^{-35} \text{ m}$,
This is called the plank length.

$$\underline{10^{-35} \text{ m}}$$

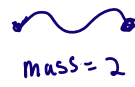
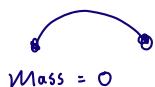
String Theory says all the particles we think are points are actually small strings that make sense when they are the size of plank length.

●
↑
electron.

They can be open or closed.



The more they vibrate the heavier they are.

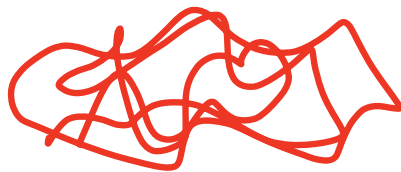


ect.

As a black hole strikes radiating particles (strings), its event horizon reaches the size of about the plank length and it just turns to a highly

Vibrating

String :



(or in your concept
a very heavy particle)

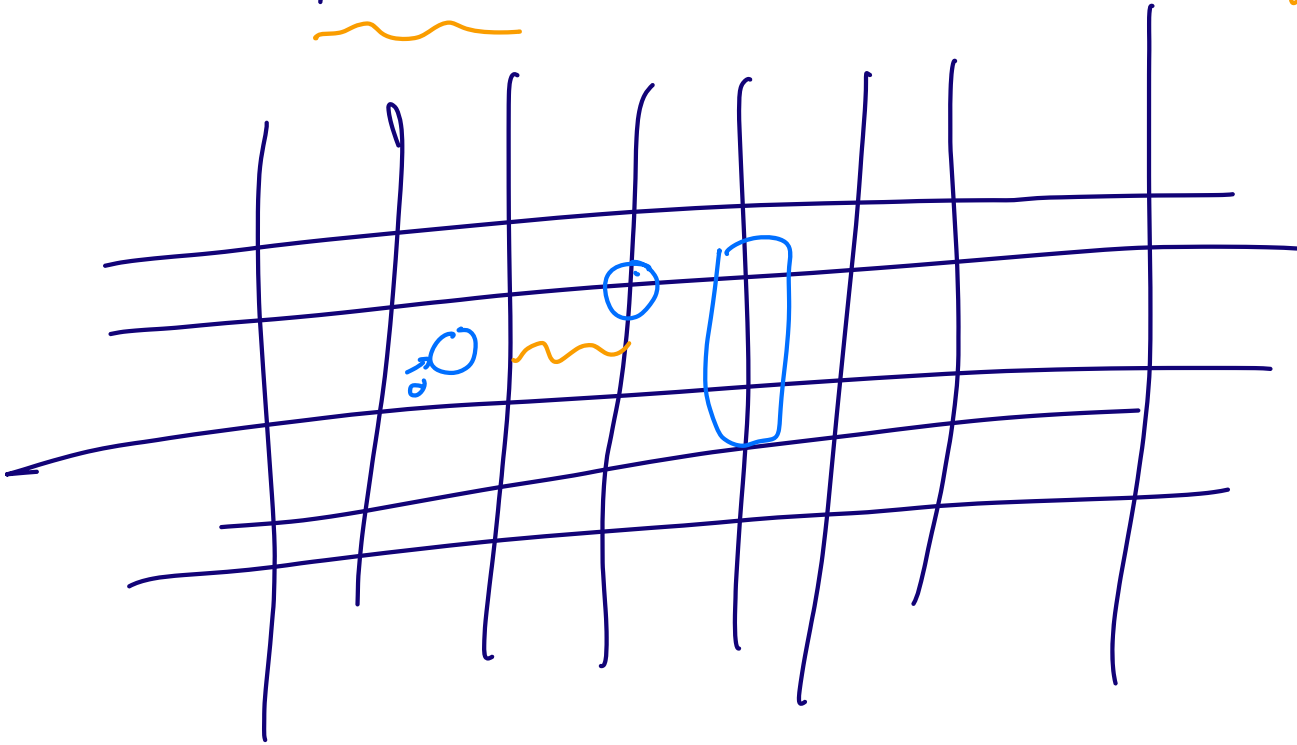
Aside:

String Theory also suggests the universe is a hologram. Just like how the event horizon is a 2D sphere covering a 3D black hole. Our 3D space might be governed by a 2D surface surrounding our universe

Check out "The holographic principle" on youtube or online

$$= \underline{10^{-35} \text{ m}} \leftarrow 10^{-22}$$

loop quantum gravity



$$O^i \quad 10^{-9} \text{ m} \quad s^1 \quad s^2$$

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad \leftarrow \text{quantum}$$

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

Brian
Green

Nicolas Gisin

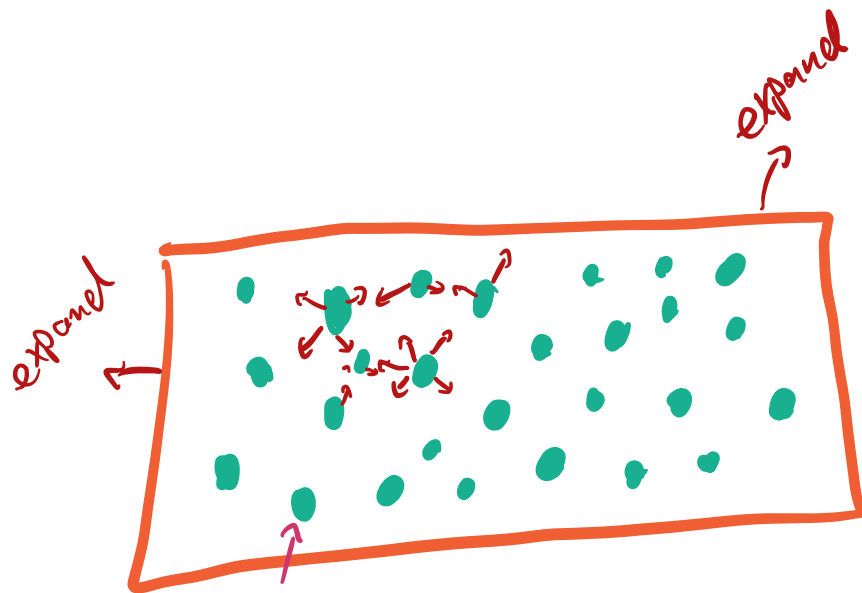
Sean
Carroll

Quantum Chance

Shape of our Universe

Recall that we can model the universe expansion by:

$$\Delta S^2 = a(t) (\Delta x^2 + \Delta y^2 + \Delta z^2) - c^2 \Delta t^2$$



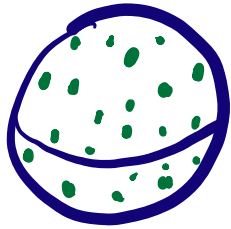
like a box
of gas

Instead of the
points being
atoms, they
are galaxies
all move away
from each other.

1 of about
2,000,000,000,000 galaxies
in the universe

The universe I have drawn, $\&$ most likely
our universe
is flat but the universe might also
be **Positively** $\&$ **Negatively** curved

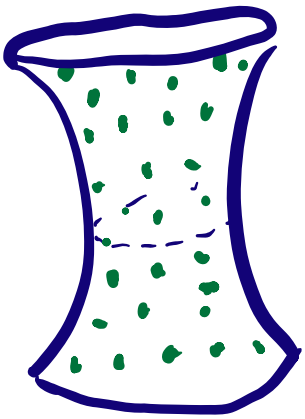
1.) Positive curvature (closed)



$\Leftarrow$ 3D ball in 4 Dimensions
(spherical)

Sum of angles > 180

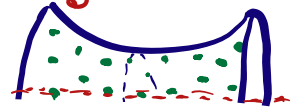
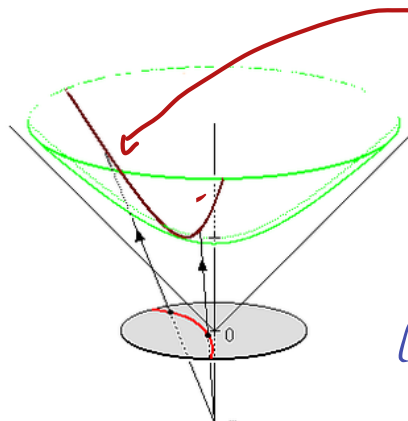
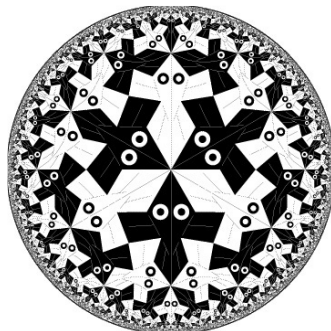
2.) Negative curvature (open)



$\Leftarrow$ hyperbolic space

Euler likes drawing hyperbolic geometry

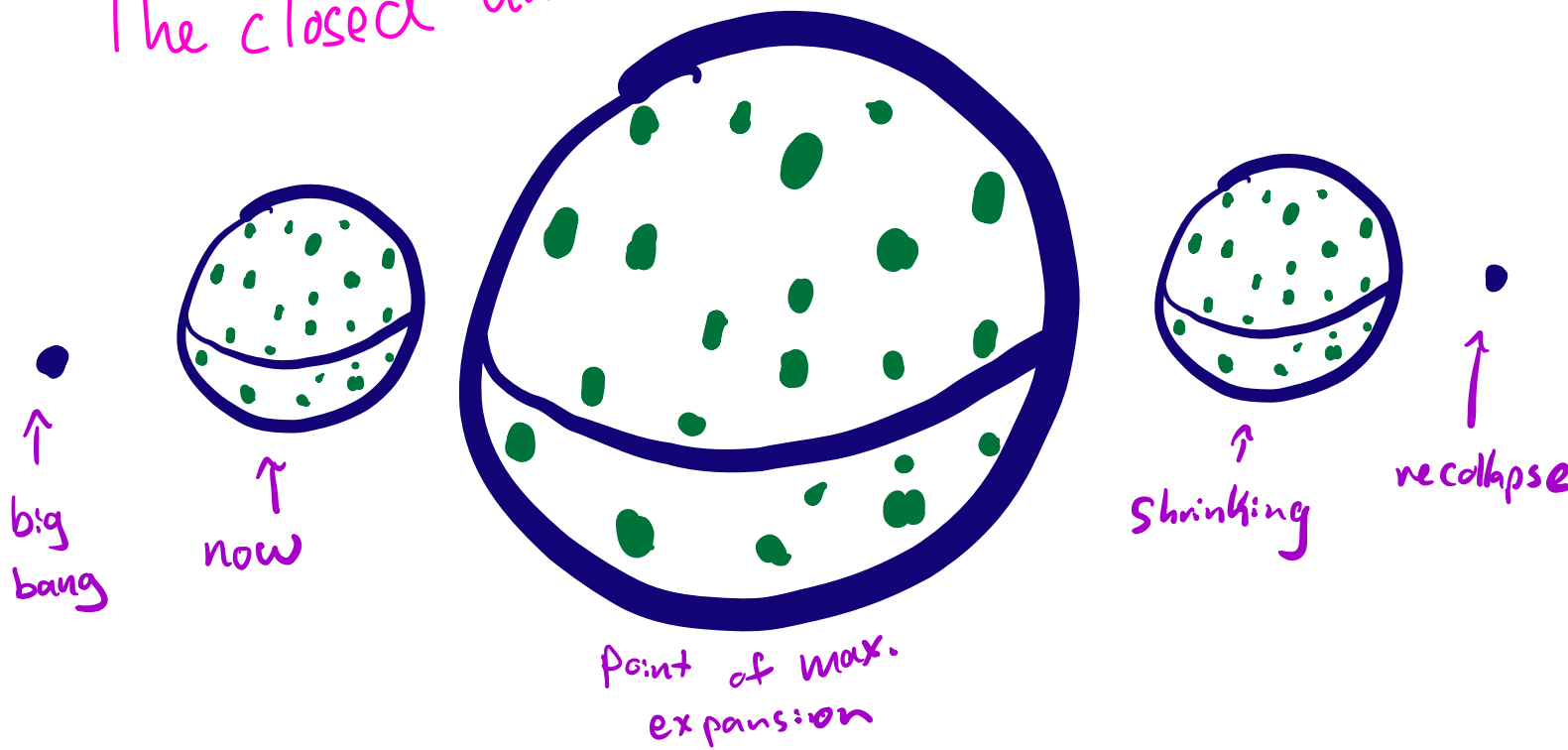
$\Leftarrow$ triangles on here will be
sum of angle $< 180^\circ$



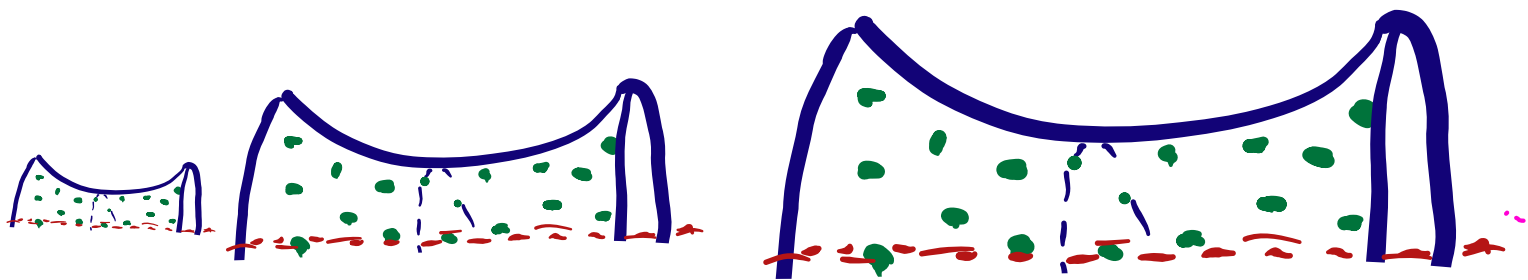
Poincaré disk
look up!

They both expand!

The closed universe recollapses



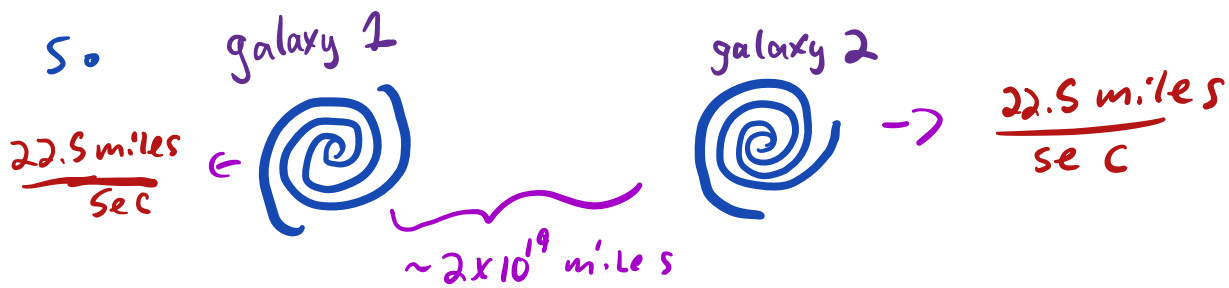
The open universe expands forever



The expansion of our universe might be caused by dark energy.

It is widely believed that this exists and can be measured by hubble variable

$$H = \frac{\text{speed of recession}}{\text{distance}} = \frac{45 \text{ miles/sec}}{1 \text{ mpc} = 1.48 \times 10^{14} \text{ miles}}$$



This is used to define a critical matter density

$$\rho_c = \frac{3H^2}{8\pi G}$$

to determine shape of our universe

Closed

$$\frac{\rho_0}{\rho_c} > 1$$

flat

$$\frac{\rho_0}{\rho_c} = 1$$

Open

$$\frac{\rho_0}{\rho_c} < 1$$

ρ_0 is matter density of our universe

The mathematical difference can be seen easily by converting this

$$\Delta s^2 = a(t) (\Delta x^2 + \Delta y^2 + \Delta z^2) - c^2 \Delta t^2$$

into polar coordinates:

$$\Delta s^2 = a^2(t) [\Delta r^2 + r^2 (\Delta \theta^2 + \sin^2 \theta \Delta \phi^2)] - c^2 \Delta t^2$$

To make more clear

open: $\Delta s^2 = a^2(t) [\Delta r^2 + \sin^2 r (\Delta \theta^2 + \sin^2 \theta \Delta \phi^2)]$

flat: $\Delta s^2 = a^2(t) [\Delta r^2 + r^2 (\Delta \theta^2 + \sin^2 \theta \Delta \phi^2)]$

closed: $\Delta s^2 = a^2(t) [\Delta r^2 + \sinh^2 r (\Delta \theta^2 + \sin^2 \theta \Delta \phi^2)]$

